(a) Level 1: decide x_{00}
0
1
2
3
0 1 2 3
UP falsifies row 0, column 0
and the main diagonal
(b) Level 2: decide x_{12}
conflict
row 2 has no free cell:
its ALO clause is violated
(c) Learn, backjump to level 1
conflict
learned clause forces $\neg x_{12}$;
UP forces x_{13}, x_{21}; row 3 empty
(d) Learn ($\neg x_{00}$), level 0
search continues and finds
a solution